12 Container X contains neon and container Y contains argon. The two containers are identical, and the two gases are at the same temperature. The pressure in X is twice that in Y.

What is the ratio of the mean kinetic energy of a neon molecule to the mean kinetic energy of an argon molecule?
[The relative atomic masses of neon and argon are 20 and 40 respectively.]

A 0.5 **B** 1 **C** 2 **D** 4

Ans: B

$$\frac{E_{k,X}}{E_{k,Y}} = \frac{\frac{3}{2}kT_X}{\frac{3}{2}kT_Y} = \frac{T_X}{T_Y} = 1$$

13 The density of helium at 150 kPa is 0.178 kg m^{-3}. What is the root-mean-square speed of its